

RVT-Swarm: Recoverability-Verified Topological Control for Decentralized Swarm Formation

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Abstract—Decentralized swarm formation in cluttered environments requires more than instantaneous collision avoidance: the team must also preserve the ability to reorganize and recover a usable formation after passing through bottlenecks. We present RVT-Swarm, a graph-based controller that augments per-robot acceleration commands with discrete topology actions and a learned *recoverability* score map over candidate formations. Recoverability is defined as a short-horizon ability to remain collision-free, make goal progress, and return toward a formation tube under a candidate topology. RVT-Swarm combines (i) a local interaction-graph encoder with action, recoverability, uncertainty, and topology heads; (ii) a counterfactual topology selector whose switching margins are derived from state and environment scale; and (iii) an optional lightweight CBF-inspired projection shield. Training uses imitation learning plus counterfactual short-horizon rollouts from a common expert controller. On the current prototype benchmark (400 evaluation episodes per method, single seed), RVT-Swarm attains the strongest learned-method success at 43.0%, together with 48.0% formation satisfaction, 66.0% collision-free execution, and the lowest stall (2.35%) and deadlock (8.25%) rates among learned controllers. The ablation study now shows that topology adaptation and counterfactual topology selection both improve over their removals, whereas the scalar recoverability path and the lightweight progress shield are neutral in this run. We therefore frame RVT-Swarm as an artifact-faithful research scaffold for recoverability-aware swarm formation control whose topology/counterfactual stack is now benchmark-positive, but whose auxiliary safety signals still require stronger activation and multi-seed validation.

Index Terms—Multi-robot systems, swarm formation control, graph neural networks, safety-critical control, topology adaptation, recoverability.

I. INTRODUCTION

Multi-robot formations are useful only when they remain both *safe* and *task-capable*. In warehouse logistics, cooperative inspection, or search-and-rescue, a team must not merely avoid collisions; it must also preserve enough structure to pass through clutter, regroup, and continue making progress toward a collective goal [1], [2], [3]. This is especially difficult in decentralized settings with limited local sensing, heterogeneous local contexts, and team sizes that vary at deployment time.

Existing approaches usually solve only part of this problem. Graph-based decentralized controllers scale well to large teams but often optimize local motion directly, without an explicit notion of whether the team can still *recover* a useful formation after local reconfiguration [4], [5]. Control-barrier-function (CBF) approaches provide strong instantaneous safety

structure [6], [7], [8], but they typically reason about collision avoidance rather than return to a formation tube. Adaptive-formation approaches can change shape to fit the environment, but usually rely on hand-designed switching logic or implicit policies without an explicit recoverability score over alternative topologies [2], [3]. Reactive methods such as ORCA can maintain goal progress, but often do so by sacrificing formation quality entirely [9].

This paper explores a different angle: rather than asking only whether the current action is safe, we ask whether the current local state is *recoverable* under a small set of candidate formation topologies. In RVT-Swarm, each local graph observation is mapped not only to a control action but also to a score map over topological actions (KEEP, COMPRESS, LINE, SPLIT, RECOVER). The controller then evaluates these topologies counterfactually and selects the one with the best predicted recoverability, subject to state-derived switching margins. An optional lightweight projection-based shield intervenes only when risk is high or when all candidate topologies appear unrecoverable.

The resulting system is deliberately pragmatic. It does not claim an exact viability kernel or a formal distributed reachability proof. Instead, it uses short-horizon counterfactual rollouts to construct a learned surrogate of recoverability and then deploys that surrogate online for topology adaptation. This design is consistent with the current codebase and the current benchmark setting.

A. Contributions

We make the following contributions:

- 1) We formulate *recoverability-aware topology adaptation* for decentralized swarm formation, where candidate formation modes are scored by their short-horizon ability to remain safe, make goal progress, and return toward a formation tube.
- 2) We implement a graph-attention controller with four coupled outputs: per-robot accelerations, graph-level topology logits, per-topology recoverability scores, and uncertainty estimates. A gradient-scaling mechanism reduces interference from auxiliary heads on the action pathway.
- 3) We introduce a recoverability-aware topology selector with state-derived switching margins, and study an optional lightweight progress-preserving projection shield that activates only under elevated risk or global negative recoverability.

- 4) We provide a unified experimental scaffold with learned baselines, historical baselines, rollout-based labels, matched-seed evaluation, and liveness metrics such as deadlock, topology switching, formation recovery time, and irreversible collapse. The released artifact exposes both the gains and the current failure modes of recoverability-aware topology control.

II. RELATED WORK

A. Decentralized Graph Policies

Graph neural networks are a natural fit for decentralized multi-robot control because they preserve permutation invariance and local communication structure [4], [5]. Our work follows this line in using a local interaction graph, but adds explicit topology adaptation and a recoverability score map rather than predicting control alone.

B. Safety-Critical Multi-Robot Control

Recent safety-critical multi-robot controllers use graph-structured certificates, collaborative safety filters, or discrete safe-policy learning to enforce pairwise constraints in distributed teams [6], [7], [8]. Recent graph-CBF work scales safety reasoning through local graphs [6], [8]. RVT-Swarm is influenced by this design, but our shield is intentionally lightweight and approximate: it is a projection-based correction mechanism rather than a formally certified QP solver. More importantly, our main decision variable is not only the continuous action but also the topology under which the swarm remains recoverable.

C. Adaptive Formation and Recoverability

Adaptive formation methods change the team geometry to fit corridors, obstacles, or task context [2], [3]. Viability and reachability learning study whether a state can remain in or return to a desirable set over a finite horizon [10], [11], [12]. RVT-Swarm sits between these directions: it learns a short-horizon, local-graph surrogate of recoverability and uses it to guide topology changes. The current implementation should therefore be understood as a recoverability-informed formation controller rather than as a formal reachability engine.

To make this positioning explicit, Table I contrasts RVT-Swarm with representative prior work along the axes that matter most for the current prototype: local graph execution, formation awareness, explicit topology adaptation, recoverability modeling, counterfactual selection, and progress-preserving safety intervention.

III. PROBLEM SETUP

A. Environment and Dynamics

We consider N robots moving in a 2D workspace with discrete-time kinematics

$$\mathbf{p}_i(t+1) = \mathbf{p}_i(t) + \mathbf{v}_i(t)\Delta t, \quad \mathbf{v}_i(t+1) = \mathbf{v}_i(t) + \mathbf{u}_i(t)\Delta t, \quad (1)$$

subject to bounded speed and acceleration. Each robot observes local relative geometry, obstacle context, goal direction, formation error, and a 36-ray 270° LiDAR scan. The

environment includes static obstacles, dynamic obstacles, and a narrow-passage configuration.

B. Formation Tube and Topology Actions

Let $\bar{\mathbf{p}}(t)$ denote the swarm centroid and let $\{\delta_i^\tau\}_{i=1}^N$ denote the desired offsets of topology $\tau \in \mathcal{T}$. The current formation error is

$$\mathbf{e}_i^\tau(t) = \bar{\mathbf{p}}(t) + \delta_i^\tau - \mathbf{p}_i(t). \quad (2)$$

Rather than choosing only a continuous control input, the policy also selects a discrete *formation mode* (or topology action) from a small finite set:

$$\mathcal{T} = \{\text{KEEP}, \text{COMPRESS}, \text{LINE}, \text{SPLIT}, \text{RECOVER}\}. \quad (3)$$

Intuitively, these modes tell the swarm whether to keep its current shape, compress into a tighter pattern, align into a line, temporarily split into subgroups, or recover back toward the nominal formation. These actions control the global formation geometry through scale, corridor alignment, or temporary splitting/merging. In the released simulator, topology actions act through four latent state variables: `topology_mode`, `formation_scale`, `split_active`, and `subteam_ids`. COMPRESS contracts the nominal spacing, LINE aligns desired offsets with the inferred corridor, SPLIT partitions robots into two laterally separated lanes, RECOVER expands the formation and clears split state, and KEEP either preserves the current corridor-aligned pattern inside a bottleneck or relaxes the swarm back toward the nominal grid outside the bottleneck.

C. Recoverability

For a state s_t and topology τ , we define recoverability as a short-horizon score that rewards collision-free motion, goal progress, and return toward a formation tube while penalizing deadlock, excessive switching, and persistent split states:

$$R(s_t; \tau) = \frac{1}{H} \sum_{h=0}^{H-1} \left[w_c c_{t+h} + w_p p_{t+h} + w_f f_{t+h} - w_d d_{t+h} - w_s q_{t+h} \right]. \quad (4)$$

Here c_{t+h} is a collision-free indicator, p_{t+h} measures goal progress, f_{t+h} measures formation-tube recovery, d_{t+h} is a deadlock-related penalty, and q_{t+h} captures switching/splitting penalties. In the implementation, these scores are generated by short-horizon counterfactual rollouts of a common expert controller. The released implementation fixes these terms once in normalized units and reuses them across all experiments; the paper intentionally presents only the score structure rather than coefficient-level artifact values.

The online objective is then to select $(\mathbf{u}, \tau)$ that preserves progress while favoring positive recoverability:

$$\max_{\mathbf{u}, \tau} \alpha \text{progress} + \beta \text{formation} + \gamma \hat{R}(s_t; \tau), \quad (5)$$

where $\hat{R}$ is the learned recoverability surrogate.

TABLE I: Positioning of RVT-Swarm relative to representative recent prior work. ✓ = primary capability; △ = partially addressed; ✗ = not addressed. The table is intended as a qualitative positioning summary for the current prototype rather than an exhaustive taxonomy.

Method	Venue	Year	Local graph distributed	Formation aware	Obstacle dynamics	Topology adaptation	Recoverability model	Counterfactual selection	Progress shield
GCBF+ [6]	T-RO	2025	✓	△	✓	✗	✗	✗	△
Collab. safety form. [7]	arXiv	2024	✓	✓	△	✗	✗	✗	△
Neural barrier reach [10]	IFAC	2024	✗	✗	△	✗	✓	✗	✗
Cert. Reach. [11]	RA-L	2025	△	✗	✓	✗	✓	✗	✗
RAIL [12]	ICRA	2025	△	✗	✓	✗	✓	✗	✗
DGPPO [8]	ICLR	2025	✓	✗	✓	✗	✗	✗	✓
Form.-adapt nav. [2]	arXiv	2024	✓	✓	△	✓	✗	✗	✗
Multi-UAV RL [3]	IROS	2025	✓	✓	✓	△	✗	✗	✗
RVT-Swarm (ours)	—	2026	✓	✓	✓	✓	✓	✓	✓

IV. METHOD

RVT-Swarm combines a graph-attention backbone, a recoverability-aware multi-head decoder, a counterfactual topology selector, and a projection-based shield. Figure 1 shows the overall pipeline and Figure 2 details the recoverability network.

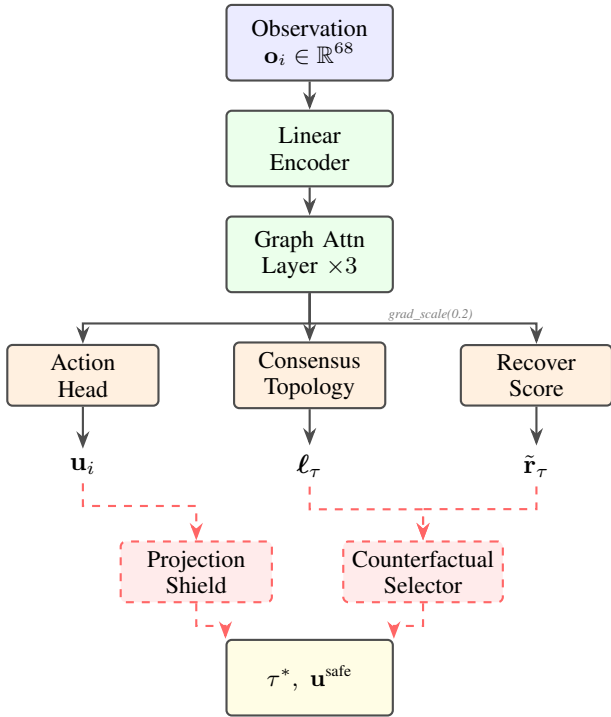


Fig. 1: Overview of RVT-Swarm. Local graph observations feed a shared graph-attention encoder whose outputs are decoded into control actions, topology logits, and per-topology recoverability scores. A counterfactual selector and a lightweight projection shield operate only at inference time.

A. Local Graph Encoder

At each step, the swarm builds a k -nearest-neighbor interaction graph with $k = 6$. Each node contains a 68-dimensional feature vector including position, velocity, heading, goal vector, centroid-relative position, formation error, obstacle-relative cues, topology encoding, normalized minimum TTC proxy,

and 36 normalized LiDAR ranges. Each edge contains 11-dimensional relational features, including relative position, relative velocity, spacing error, corridor projections, and a TTC-style safety descriptor.

Table II summarizes the exact feature decomposition used by the released encoder. The total dimensions are 68 per node and 11 per directed edge.

TABLE II: Node and edge feature blocks for the local interaction graph.

Scope	Block	Symbols	Dim.
Node	Kinematics	$\mathbf{p}_i, \mathbf{v}_i, \text{head}_i$	6
Node	Goal cues	$\mathbf{g}_i, \hat{\mathbf{g}}_i$	4
Node	Formation context	$(\mathbf{p}_i - \bar{\mathbf{p}}), \mathbf{e}_i^\tau$	4
Node	Obstacle/corridor	$(\mathbf{p}_i - \mathbf{p}_{\text{obs},c}), \Pi_i^{\text{corr}}, \Pi_i^{\text{lat}}$	4
Node	Global state	$\kappa_t, b_t, \pi_t, \chi_t, \text{onehot}(m_t)$	9
Node	Local safety cues	$\rho_i^{\text{local}}, d_i^{\text{min}}, \text{TTC}_i^{\text{min}}$	3
Node	Dynamic-obstacle cue	$\mathbf{v}_i^{\text{dyn}}$	2
Node	LiDAR scan	$\text{LiDAR}_{i,1:36}$	36
Edge	Relative geometry	$\Delta \mathbf{p}_{ij}, \ \Delta \mathbf{p}_{ij}\ $	3
Edge	Relative velocity	$\Delta \mathbf{v}_{ij}$	2
Edge	Corridor/spacing	$\Pi_{ij}^{\text{corr}}, \Pi_{ij}^{\text{lat}}, \epsilon_{ij}^{\text{space}}$	3
Edge	Safety/global context	$\text{TTC}_{ij}, b_t, \pi_t$	3

For completeness, the exact concatenated node and edge vectors are

$$\mathbf{x}_i = [\mathbf{p}_i, \mathbf{v}_i, \text{head}_i, \mathbf{g}_i, \hat{\mathbf{g}}_i, (\mathbf{p}_i - \bar{\mathbf{p}}), \mathbf{e}_i^\tau, (\mathbf{p}_i - \mathbf{p}_{\text{obs},c}), \Pi_i^{\text{corr}}, \Pi_i^{\text{lat}}, \kappa_t, b_t, \pi_t, \rho_i^{\text{local}}, d_i^{\text{min}}, \mathbf{v}_i^{\text{dyn}}, \chi_t, \text{onehot}(m_t), \text{TTC}_i^{\text{min}}, \text{LiDAR}_{i,1:36}], \quad (6)$$

and

$$\mathbf{e}_{ij} = [\Delta \mathbf{p}_{ij}, \Delta \mathbf{v}_{ij}, \|\Delta \mathbf{p}_{ij}\|, \Pi_{ij}^{\text{corr}}, \Pi_{ij}^{\text{lat}}, \epsilon_{ij}^{\text{space}}, \text{TTC}_{ij}, b_t, \pi_t]. \quad (7)$$

Here $\text{head}_i = (\cos \theta_i, \sin \theta_i)$ is a velocity-derived heading encoding, $\mathbf{g}_i$ and $\hat{\mathbf{g}}_i$ are raw and normalized goal vectors, $\mathbf{p}_{\text{obs},c}$ is obstacle centroid, Π_i^{corr} and Π_i^{lat} denote projections onto the inferred corridor and lateral axes, and m_t is the current topology mode. Both $\text{TTC}_i^{\text{min}}$ and TTC_{ij} are computed from the quadratic collision equation for circular agents rather than from a purely geometric distance threshold. The graph is directed because each robot independently selects its own six nearest neighbors.

The graph is processed by three attention-weighted message-passing layers. For edge $(j \rightarrow i)$, an edge MLP

produces a message $\mathbf{m}_{ij}$ and a learned attention weight α_{ij} , and the node update is

$$\mathbf{m}_{ij} = \phi_e([\mathbf{h}_i \parallel \mathbf{h}_j \parallel \mathbf{e}_{ij}]), \quad (8)$$

$$\alpha_{ij} = \text{softmax}_{j \in \mathcal{N}(i)}(\phi_a(\mathbf{m}_{ij})), \quad (9)$$

$$\mathbf{h}'_i = \mathbf{h}_i + \phi_n\left([\mathbf{h}_i \parallel \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{m}_{ij}]\right). \quad (10)$$

The hidden dimension is 128 throughout. In the released model, an encoder MLP first maps $68 \rightarrow 128 \rightarrow 128$, and each message-passing layer uses an edge MLP on $[\mathbf{h}_j \parallel \mathbf{h}_i \parallel \mathbf{e}_{ij}]$ followed by a per-destination softmax over incoming messages. Because k is fixed, the encoder cost scales approximately linearly with swarm size, $\mathcal{O}(Nkd)$, which is one reason the policy can be evaluated on different team sizes without changing the architecture.

B. Recoverability Field Network

The shared backbone feeds four decoders:

- an **action head** producing per-robot accelerations $\mathbf{u}_i \in \mathbb{R}^2$;
- a **topology consensus head** producing graph-level logits over $\mathcal{T}$;
- a **score head** producing per-topology recoverability scores;
- an **uncertainty head** producing a nonnegative uncertainty estimate for each candidate topology.

The topology head is not a simple pooled classifier. Each node first casts a topology vote, neighbor votes are aggregated into mean and spread statistics, and a small agreement MLP refines these votes before graph-level pooling. This gives the model a limited form of local consensus without requiring explicit communication optimization. The consensus path is summarized in Algorithm 1.

Algorithm 1: Neighborhood Topology Consensus

Input: Node embeddings $\{\mathbf{h}_i\}$ and graph neighborhood $\mathcal{N}(i)$

Output: Graph-level topology logits ℓ

- 1 $\mathbf{v}_i \leftarrow \text{MLP}_{\text{vote}}(\mathbf{h}_i) \quad \forall i$;
 - 2 $\bar{\mathbf{v}}_i \leftarrow \text{MeanAgg}_{j \in \mathcal{N}(i)}(\mathbf{v}_j)$;
 - 3 $\sigma_i \leftarrow \sqrt{\text{VarAgg}_{j \in \mathcal{N}(i)}(\mathbf{v}_j) + \epsilon}$;
 - 4 $\hat{\mathbf{v}}_i \leftarrow \text{MLP}_{\text{agree}}([\mathbf{v}_i \parallel \bar{\mathbf{v}}_i \parallel \sigma_i])$;
 - 5 $\ell \leftarrow \text{MeanPool}(\hat{\mathbf{v}}_i)$;
 - 6 **return** ℓ
-

The four heads operate at two different granularities. Actions are robot-wise, whereas topology choice and recoverability are graph-wise. Let $\mathbf{h}_i^{\text{aux}} = \text{GradScale}(\mathbf{h}_i; \gamma_{\text{aux}})$ and $\bar{\mathbf{h}} = \frac{1}{N} \sum_i \mathbf{h}_i^{\text{aux}}$. The released implementation computes

$$\begin{aligned} \mathbf{u}_i &= \tanh(\phi_{\text{act}}(\mathbf{h}_i)), \\ \hat{\mathbf{r}} &= \phi_{\text{score}}(\bar{\mathbf{h}}), \\ \hat{\sigma} &= \text{softplus}(\phi_{\text{unc}}(\bar{\mathbf{h}})), \\ \hat{\mathbf{a}} &= \phi_{\text{aux}}(\bar{\mathbf{h}}), \end{aligned} \quad (11)$$

where the auxiliary head predicts four graph-level scalars: formation scale, bottleneck score, normalized progress, and

split activity. This separation is deliberate: the continuous action is local and per-robot, while topology and recoverability are properties of the whole team configuration.

Recoverability scores are adjusted by dispersion-scaled uncertainty:

$$\tilde{r}_\tau = \hat{r}_\tau - \sigma(\hat{\mathbf{r}}) \hat{\sigma}_\tau, \quad \hat{R}(s_t) = \max_{\tau \in \mathcal{T}} \tilde{r}_\tau. \quad (12)$$

Only the action head receives full-strength gradients from the backbone. Before feeding the auxiliary heads, we apply gradient scaling with $\gamma_{\text{aux}} = 1/(K + 1)$, where K is the number of message passing rounds, so the auxiliary coupling is tied to backbone depth rather than treated as a separately tuned coefficient. The action head is slightly deeper than the graph-level heads, while the topology-vote, agreement, score, uncertainty, and auxiliary decoders all remain shallow shared-width MLPs. We intentionally omit release-level layer widths from the main text because they are implementation detail rather than an algorithmic claim.

C. Counterfactual Topology Selection

At inference, RVT-Swarm evaluates all candidate topologies using a combination of uncertainty-adjusted recoverability, topology prior, contextual bonuses, validity masks, and switching cost:

$$\begin{aligned} C_\tau &= \pi_\tau + \phi(\tilde{r}_\tau) + b_\tau^{\text{ctx}} \\ &\quad - \sigma_\tau - \psi_\tau^{\text{switch}} + m_\tau^{\text{valid}}, \end{aligned} \quad (13)$$

The context term favors, for example, LINE or COMPRESS in bottlenecks and RECOVER once the bottleneck clears. The switch penalty discourages excessive topology changes, especially disruptive split actions. The current selector is a two-stage gate rather than a single argmax: it starts from the topology logit winner, permits a recoverability-based override only when the score gain exceeds an uncertainty-dependent margin, and finally applies cooldown and hysteresis checks with additional margins for specialist actions (LINE, SPLIT, RECOVER). In the released implementation, the prior temperature, switching cooldown, and hysteresis margin are not exposed as free user-tuned knobs; instead they are derived from environment scale quantities such as d_0 , $d_{\min}$, sensing range, and $v_{\max} \Delta t$. The online selection-and-shield loop is summarized in Algorithm 2.

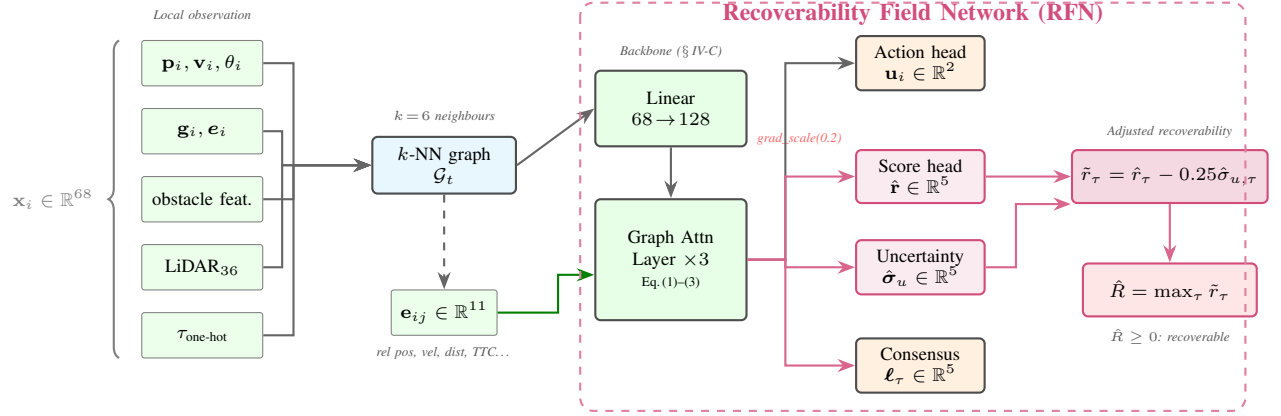


Fig. 2: Recoverability Field Network (RFN). Local node and edge features are encoded by a three-layer graph-attention backbone and decoded into action, recoverability, uncertainty, and topology outputs. The adjusted score map $\tilde{r}_\tau$ is used by the topology selector and by the shield trigger.

Algorithm 2: Counterfactual Topology Selection with Projection Shield

Input: Observation o_t , topology logits ℓ , adjusted scores $\tilde{r}$, uncertainty σ , previous topology τ_{prev} , nominal action $\mathbf{u}$

Output: Selected topology τ^* , shielded action $\mathbf{u}^{\text{safe}}$

```

1 Compute prior  $\pi = \text{softmax}(\ell/T_{\text{env}})$ ;
2 Compute contextual bonuses  $b_\tau^{\text{ctx}}$ ;
3 Compute switching penalties  $\psi_\tau^{\text{switch}}$ ;
4 Compute  $C_\tau$  using (13);
5  $\tau' \leftarrow \arg \max_{\tau \in \mathcal{T}} C_\tau$ ;
6 Compute uncertainty/specialist margin  $\eta_{\text{dyn}}$ ;
7 if  $C_{\tau'} < C_{\tau_{\text{prev}}} + \eta_{\text{dyn}}$  then
8    $\tau^* \leftarrow \tau_{\text{prev}}$ ;
9 else
10   $\tau^* \leftarrow \tau'$ ;
11 end
    // Risk-triggered projection shield
12  $\rho \leftarrow \text{CollisionRisk}(o_t)$ ;
13 if  $\max_{\tau \in \mathcal{T}} \tilde{r}_\tau < 0$  then
14    $\rho \leftarrow \max(\rho, |\max_{\tau \in \mathcal{T}} \tilde{r}_\tau|)$ ;
15 end
16 if  $\rho > 0$  then
17   Build pairwise robot-robot and robot-obstacle half-planes;
18   Project  $\mathbf{u}$  onto the feasible intersection and acceleration ball;
19   Blend projected and nominal actions using a geometry-derived blend cap;
20 end
21 else
22    $\mathbf{u}^{\text{safe}} \leftarrow \mathbf{u}$ ;
23 end
24 return  $\tau^*, \mathbf{u}^{\text{safe}}$ 

```

D. Geometry Execution of Topology Actions

The selector does not directly reposition robots; instead it updates a latent formation state consisting of the discrete

mode m_t , formation scale κ_t , split-activity variable χ_t , and optional subteam assignment vector $\mathbf{s}_t$. Desired offsets are then regenerated relative to the inferred corridor direction $\mathbf{c}_t$ and lateral axis $\mathbf{l}_t = [c_{t,y}, -c_{t,x}]^\top$.

For LINE, the desired geometry becomes a one-dimensional lattice along the corridor:

$$\delta_i^{\text{line}} = \mathbf{c}_t \left(i - \frac{N-1}{2} \right) d_0 \kappa_t. \quad (14)$$

For SPLIT, robots are sorted by lateral projection $\mathbf{p}_i^\top \mathbf{l}_t$ and divided into two subteams; each subteam occupies a lane centered at $\pm \frac{1}{2} g_t \mathbf{l}_t$ with longitudinal spacing $d_0 \kappa_t$, where g_t is a safety-aware lane gap derived from the current spacing and minimum robot-robot clearance. The nominal KEEP/COMPRESS/RECOVER modes use a grid aligned with $(\mathbf{c}_t, \mathbf{l}_t)$.

In the released simulator, topology actions act through smoothed state updates rather than abrupt geometry jumps: COMPRESS contracts κ_t toward a bottleneck-dependent target, LINE activates corridor alignment and a stronger contraction target, SPLIT activates two-lane assignments and increases χ_t , RECOVER clears split state and expands the formation, and KEEP either preserves the bottleneck configuration or drifts back toward the nominal grid. The topology-switch counter increments whenever the discrete mode changes or the induced desired spacing changes by more than a robot-radius-equivalent displacement, which is why the selector must control both categorical switches and repeated scale-only oscillations.

E. Progress-Preserving Projection Shield

The shield activates only when collision risk is high or when all candidate topologies have negative recoverability. Pairwise robot-robot and robot-obstacle constraints are converted into half-plane constraints in action space using a CBF-inspired linearization. For robot i , we use the barrier

$$h_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|^2 - d_{\text{safe}}^2 \quad (15)$$

and enforce

$$\nabla h_{ij}^\top \mathbf{u}_i \geq -\alpha h_{ij} - 2(\mathbf{p}_i - \mathbf{p}_j)^\top (\mathbf{v}_i - \mathbf{v}_j). \quad (16)$$

Rather than solving a full external QP, we iteratively project a progress-biased target action onto the intersection of these half-planes and the acceleration ball. The final shielded action is softly blended with the learned action, which reduces overreaction under moderate risk. Recoverability is used conservatively: positive recoverability does not suppress geometric collision risk, but globally negative recoverability raises the intervention floor and therefore makes the shield more willing to intervene during unrecoverable crises. In the released artifact, the trigger level and blend cap are derived from environment scale quantities such as nominal spacing, clearance, and $v_{\max} \Delta t$, rather than being exposed as hand-tuned runtime thresholds.

V. TRAINING

A. Data Generation

Training data are generated from a common heuristic expert controller in the shared simulator. For each sampled state, we run short-horizon counterfactual rollouts for every topology in $\mathcal{T}$ using the same expert controller, producing:

- a per-topology score vector;
- a best-topology label;
- a scalar recoverability margin derived from the score gap and the best score.

The current release uses 500 episodes spanning four scenarios (*open field*, *cluttered*, *narrow passage*, *dynamic obstacles*) and team sizes $N \in \{4, 8, 12, 16\}$. Scenarios are sampled from one shared scenario pool, and hard-negative mining augments low-margin constrained states using perturbations whose amplitude grows with recoverability deficit.

For artifact fidelity, the shared expert itself is not an abstract oracle but a deterministic scale-normalized controller used everywhere in the release for both label generation and counterfactual rollouts. It combines goal seeking, topology-conditioned formation correction, velocity damping, and local robot-robot / robot-obstacle repulsion with TTC-style urgency, all normalized by environment scales before clipping to the acceleration limit. The same structure is reused across training, ablation, and evaluation runs without per-scenario retuning.

The counterfactual rollout score that defines $\mathbf{q}(s_t)$ is also computed directly from the simulator rather than learned. In practice, it aggregates normalized collision-free execution, goal progress, formation-tube recovery, switching cost, deadlock/stall penalties, and terminal success/collapse events over the horizon H through one shared normalized aggregation rule, which is reused across all experiments.

Let $\mathbf{q}(s_t) \in \mathbb{R}^{|\mathcal{T}|}$ denote the rollout-score vector for a sampled state and let $\tau^* = \arg \max_{\tau \in \mathcal{T}} q_\tau$. The released code forms the supervised targets as

$$\begin{aligned} \mathbf{y}^{\text{score}} &= \frac{\mathbf{q} - \mu(\mathbf{q})\mathbf{1}}{\max(\sigma(\mathbf{q}), \epsilon)}, \\ y^{\text{topo}} &= \tau^*, \\ y^{\text{rec}} &= \tanh(\bar{q}_{\tau^*} + [\bar{q}_{\tau^*} - \bar{q}_{(2)}] + \mathbb{I}[\tau^* = \text{RECOVER}]), \\ \mathbf{y}^{\text{act}} &= \mathbf{u}^{\text{expert}}(s_t, \tau^*) / u_{\max}, \end{aligned} \quad (17)$$

where $\mu(\mathbf{q})$ and $\sigma(\mathbf{q})$ are the mean and standard deviation of the rollout-score vector, $\mathbf{1}$ is the all-ones vector, ϵ is a small floor used to avoid division by a near-zero standard deviation, and $q_{(2)}$ is the second-best rollout score; equivalently, $\bar{\mathbf{q}}$ denotes the standardized rollout score vector. Thus the scalar recoverability target is not an independent label; it is a distilled margin derived from the topology score vector itself. In the released dataset generator, low-margin bottleneck states also receive perturbed hard negatives, again with one fixed recipe shared by all experiments.

B. Loss and Optimization

For RVT-Swarm, the training loss is the mean of the active objectives rather than a hand-balanced weighted sum:

$$\mathcal{L} = \text{Mean}(\mathcal{S}_{\text{active}}), \quad (18)$$

where

$$\bar{\mathcal{L}}_{\text{topo}} = \frac{1}{3}(\mathcal{L}_{\text{topo}} + \mathcal{L}_{\text{score}} + \mathcal{L}_{\text{rank}}) \quad (19)$$

is an equal-weight bundle of the topology-classification, score-map, and pairwise-ranking objectives. For GNN_ONLY, $\mathcal{S}_{\text{active}} = \{\mathcal{L}_{\text{act}}\}$; for INSTANT_CERT, $\mathcal{S}_{\text{active}} = \{\mathcal{L}_{\text{act}}, \mathcal{L}_{\text{rec}}\}$; and for topology-enabled RVT-Swarm, $\mathcal{S}_{\text{active}} = \{\mathcal{L}_{\text{act}}, \bar{\mathcal{L}}_{\text{topo}}, \mathcal{L}_{\text{aux}}, \mathcal{L}_{\text{unc}}\}$. Thus the release no longer depends on a separate vector of loss-balancing coefficients. The scalar recoverability regression term is retained for InstantCert but omitted for topology-enabled RVT-Swarm, because the score-map and ranking objectives were empirically more stable than jointly optimizing an additional scalar recoverability head. The ranking term is implemented as a softplus sign-consistency loss over all topology pairs with unequal target scores, and the uncertainty penalty is dispersion-normalized so that inflating uncertainty is discouraged without tying the calibration term to a hand-chosen fixed coefficient.

The three learned methods are trained with one fixed AdamW configuration shared across all reported runs; the exact release values are listed in the appendix rather than in the main text. Best-checkpoint selection is based on periodic rollout validation: starting at the first rollout-validation checkpoint, every ten epochs we evaluate a small validation set over *narrow passage* and *dynamic obstacles* for team sizes 8 and 16 with four episodes per setting, and select checkpoints using

$$\begin{aligned} \text{Score}_{\text{val}} &= \text{Mean}(\text{Success}, \text{CollFree}, \text{FormOK}, \text{Goal}) \\ &\quad - \text{Mean}(\text{Collapse}, \text{Deadlock}, \text{Stall}). \end{aligned} \quad (20)$$

This is the exact scalar used by the training script once rollout-based checkpointing activates; it is not a hand-written summary added after the fact.

VI. EXPERIMENTS

A. Experimental Protocol

The benchmark uses a common 2D simulator with workspace 12×12 m, time step $\Delta t = 0.15$ s, robot radius 0.18 m, obstacle radius 0.35 m, speed limit 0.9 m/s, acceleration limit 0.6 m/s², and a 36-ray LiDAR with 3 m range. Main evaluation uses 25 episodes per setting over 4 scenarios and 4

team sizes, for 400 episodes per method. Episodes are matched across methods by shared seeds,

$$\text{seed}(s, n, e) = s_0 + 10,000 \text{idx}(s) + 100n + e, \quad (21)$$

with $s_0 = 42$ in the current release; rollout-validation seeds use the same pattern with an additional offset of 50,000.

The primary success metric is also conjunctive in the released simulator:

$$\text{Success} = \mathbb{I}[\text{GoalReached} \wedge \text{CollisionFree} \wedge \text{FormOK}], \quad (22)$$

where `CollisionFree` requires zero robot-robot and robot-obstacle collisions over the episode, and `FormOK` = $\mathbb{I}[\text{FormRMS} < \epsilon_{\text{form}}]$. Because reviewers often ask whether these are post-processed metrics, we also give the exact released definitions of deadlock, irreversible collapse, and recoverability calibration in the appendix.

a) Implementation scope.: This benchmark is intentionally a *common-simulator prototype*. The non-learned baselines (ADAPTIVE_FORMATION, CBF_QP_LIKE, ORCA_LIKE, and CENTRALIZED_MPC) are internal normalized baselines rather than exact package-level reproductions. The current paper reports the single-seed result produced by the released code path; multi-seed confidence intervals remain future work. For artifact fidelity, we also note that the released `ms/step` field is a normalized runtime proxy from the evaluation script (1.0 for non-RVT methods and 1.6 for RVT-Swarm), not a wall-clock timing benchmark.

B. Compared Methods

GNN-Only uses the shared graph backbone with only an action head. **InstantCert** adds a scalar recoverability-style head but no topology adaptation. **RVT-Swarm** is the full model. Historical baselines include a heuristic adaptive-formation controller, a CBF-QP-like reactive filter, ORCA-like avoidance, and a centralized MPC-inspired proxy. For code-level reproducibility, these historical baselines are not left as labels:

- **ORCA-like** uses clipped goal tracking with velocity damping plus pairwise and obstacle repulsion, with no topology adaptation.
- **Adaptive formation** selects a heuristic topology from bottleneck/progress comparisons and then calls the same shared heuristic expert controller used for data generation and counterfactual rollouts.
- **CBF-QP-like** starts from the KEEP-topology expert action and adds stronger reactive repulsion terms before clipping.
- **Centralized MPC proxy** also reuses the expert controller with a heuristic topology, then adds a small global centroid-to-goal bias.

These are therefore normalized internal baselines in one simulator, not claims of package-identical external reproductions.

C. Main Results

Three points stand out from Table III. First, RVT-Swarm is now the strongest learned controller in the current artifact:

it attains the highest learned-method success, collision-free execution, formation satisfaction, and lowest formation RMS, while InstantCert remains the closest runner-up. Second, RVT-Swarm also retains the clearest liveness profile among learned controllers, with the lowest stall and deadlock rates. Together, these results indicate that the topology/counterfactual stack is no longer only preventing stagnation; in this run it also translates into better end-task completion and formation quality. Third, the historical baselines still expose the intended trade-offs: ORCA-like motion reaches the goal often but abandons formation entirely, while CBF-QP-like control maximizes collision-free execution at the cost of formation completion.

For completeness, the released artifact also logs topology switches, formation recovery time, and recoverability calibration. In the latest run, RVT-Swarm averages 0.72 topology switches and formation recovery time 204.00, compared with 0 switches and 210.43 for InstantCert. Recoverability false-positive / false-negative rates are 0.0540 / 0.0017 for RVT-Swarm versus 0.0461 / 0.0011 for InstantCert. Table VI lists the remaining per-method metrics omitted from Table III.

D. Ablation Results

TABLE IV: Ablation results for RVT-Swarm. In the current run, topology adaptation and counterfactual selection provide the clearest gains, while the scalar recoverability and progress-shield toggles are effectively neutral.

Variant	Success $\uparrow$	Coll-Free $\uparrow$	Form OK $\uparrow$	Deadlock $\downarrow$	Collapse $\downarrow$	Topo Sw. $\downarrow$
Full	0.430	0.660	0.480	0.083	0.123	0.720
– Recoverability	0.430	0.660	0.480	0.083	0.123	0.720
– Counterfactual	0.373	0.640	0.415	0.060	0.135	1.195
– Progress shield	0.430	0.660	0.480	0.083	0.123	0.720
– Topology	0.378	0.668	0.423	0.075	0.100	0.000
Fixed formation	0.378	0.668	0.423	0.075	0.100	0.000

The new ablation table is substantially cleaner than the earlier prototype because the main RVT-Swarm result and the ablation *full* row now match exactly, eliminating the previous checkpoint-selection mismatch. Two effects are now unambiguous. Removing counterfactual selection drops success from 0.430 to 0.373 and increases topology switches from 0.72 to 1.20, showing that the counterfactual selector is the main source of topology stability. Removing topology adaptation altogether also hurts the end task, reducing success and formation satisfaction to 0.378 and 0.423 even though raw collision-free execution improves slightly. At the same time, *no-recoverability* and *no-progress-shield* are numerically identical to the full model in this single-seed run, indicating that those two paths are currently neutral under the present evaluation distribution. As before, *fixed formation* and *no-topology* remain nearly degenerate once topology execution is disabled, so their gap should not be over-interpreted. Table VII reports the remaining ablation metrics omitted from Table IV.

E. Interpretation

The current results suggest a more specific interpretation. Relative to InstantCert and GNN-Only, RVT-Swarm is no longer only a liveness-oriented variant; in this run it is the best learned controller on success, collision-free execution,

TABLE III: Main benchmark results averaged over all scenarios and team sizes. Numbers are from the current single-seed prototype run. The final column is a normalized runtime proxy from the released evaluation script, not a hardware-timed latency measurement.

Method	Success $\uparrow$	Goal $\uparrow$	Coll-Free $\uparrow$	Form OK $\uparrow$	Form RMS $\downarrow$	Stall $\downarrow$	Deadlock $\downarrow$	Collapse $\downarrow$	ms/step $\downarrow$
ORCA-like	0.000	0.920	0.633	0.000	2.907	0.005	0.010	0.018	1.0
Adaptive Formation	0.303	0.778	0.563	0.358	0.972	0.029	0.088	0.143	1.0
CBF-QP-like	0.183	0.753	0.723	0.205	0.847	0.027	0.080	0.085	1.0
Centralized MPC	0.318	0.763	0.603	0.345	0.963	0.030	0.105	0.163	1.0
GNN-Only	0.378	0.790	0.628	0.428	0.789	0.032	0.098	0.135	1.0
InstantCert	0.408	0.763	0.633	0.453	0.760	0.033	0.113	0.135	1.0
RVT-Swarm	0.430	0.795	0.660	0.480	0.758	0.023	0.083	0.123	1.6

and formation quality, while still keeping the lowest stall and deadlock. The main residual gap is between goal reaching (79.5%) and strict formation satisfaction (48.0%), indicating that the harder part is maintaining or recovering formation after making progress, not navigation itself. The ablation table further localizes the useful mechanisms: counterfactual selection and topology adaptation now carry the measurable gain, whereas the scalar recoverability path and the progress shield do not alter the aggregate behavior on this benchmark. This is consistent with the current codebase, where topology choice is driven primarily by the per-topology score map and handcrafted context plus scale-derived switching logic, while the scalar recoverability objective remains disabled during RVT training.

VII. LIMITATIONS AND FUTURE WORK

This work should be read as a research prototype with promising algorithmic structure, not as a finished benchmark artifact. The main limitations are:

- **Short-horizon recoverability surrogate.** Recoverability labels are derived from a heuristic expert over a finite horizon rather than from a globally optimal planner or a formally certified reachability computation.
- **Prototype baselines.** The historical baselines are normalized in-house implementations in a common simulator, not exact reproductions of external packages.
- **Single-seed reporting.** The current paper reports one full prototype sweep. A submission intended for a top-tier venue should include multi-seed confidence intervals, statistical testing, and a more mature baseline suite.
- **Some auxiliary safety components are not yet benchmark-positive.** In the current single-seed run, removing the scalar recoverability path or the progress shield leaves the aggregate metrics unchanged, and *fixed formation / no-topology* remain nearly degenerate. This suggests the topology/counterfactual stack is active, but parts of the auxiliary safety machinery are still under-exercised.
- **Runtime proxy, not timing benchmark.** The released `ms/step` field is a normalized proxy baked into the evaluation script, not a synchronized wall-clock latency study.
- **Simplified shield and dynamics.** The shield uses iterative half-plane projection, and the environment is 2D with simplified kinematics. Extensions to nonholonomic and 3D platforms remain open.

• Portability still needs explicit robustness evidence.

Although the released defaults are fixed once and reused across all runs rather than re-tuned per scenario, the current paper does not yet report cross-environment sensitivity or transfer sweeps. A stronger version should show that the normalized score structure and selector logic remain stable under changed workspace scale, obstacle density, sensing range, and actuation limits.

The clearest next steps are to (i) design evaluation slices that actually activate the scalar recoverability and shield paths, (ii) strengthen formation-maintenance and recovery objectives after goal progress, and (iii) evaluate the method over multiple seeds, transfer perturbations, and richer baseline reproductions.

VIII. CONCLUSION

We presented RVT-Swarm, a decentralized graph-based controller that combines continuous control, discrete topology adaptation, and a learned recoverability score map for swarm formation control in constrained environments. In the current artifact, RVT-Swarm is the strongest learned controller on success and formation quality while also reducing stall and deadlock relative to the simpler learned baselines. The ablation study now indicates that topology adaptation and counterfactual selection contribute real gains, whereas the scalar recoverability and progress-shield paths remain neutral under single-seed evaluation. The artifact therefore supports RVT-Swarm as a promising recoverability-aware formation-control scaffold with a functioning topology layer, but not yet as a fully validated auxiliary-safety stack.

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APPENDIX

APPENDIX: EXPERIMENTAL SETTINGS

TABLE V: Experimental settings reported in the paper.

Parameter	Value
<i>Environment</i>	
Workspace	12.0 × 12.0 m
Δt	0.15 s
$v_{\max} / u_{\max}$	0.9 / 0.6
Robot / obstacle radius	0.18 / 0.35
LiDAR rays / range / FOV	36 / 3.0 m / 270°
Team sizes	{4, 8, 12, 16}
Scenarios	open, cluttered, narrow, dynamic
<i>Model</i>	
Node feature dim / edge dim	68 / 11
Hidden dim / message passes	128 / 3
Graph neighborhood k	6
<i>Training</i>	
Expert episodes	500
Batch size	32
Recoverability horizon H	14
Early stopping patience	40
RVT max epochs	300
GNN / InstantCert max epochs	120 / 120
<i>Rollout validation</i>	
Enabled	yes
Interval	every 10 epochs
Scenarios	narrow passage, dynamic obstacles
Team sizes	8, 16
Episodes per setting	4
<i>Evaluation</i>	
Episodes per setting	25
Runtime field	normalized proxy, not wall-clock

Note: the current release exposes shared experimental settings such as environment scale, horizon, and optimizer setup, but does not expose separate user-tuned loss-balancing coefficients or selector/shield thresholds. Runtime margins are derived from the same environment-scale quantities used everywhere else in the simulator.

APPENDIX: COMPLETE SINGLE-SEED METRIC DUMPS

TABLE VI: Additional per-method metrics omitted from Table III.

Method	RR Col.↓	RO Col.↓	Topo Sw.	Rec. Time↓	R-FP↓	R-FN↓
ORCA-like	0.0002	0.0078	0.000	262.72	0.0000	0.0000
Adaptive Formation	0.0083	0.0104	0.500	229.02	0.0000	0.0000
CBF-QP-like	0.0015	0.0043	0.000	254.91	0.0000	0.0000
Centralized MPC	0.0088	0.0113	0.750	233.75	0.0000	0.0000
GNN-Only	0.0082	0.0087	0.000	209.71	0.0000	0.0000
InstantCert	0.0075	0.0084	0.000	210.43	0.0461	0.0011
RVT-Swarm	0.0093	0.0078	0.720	204.00	0.0540	0.0017

TABLE VII: Additional ablation metrics omitted from Table IV.

Variant	Goal↑	RR Col.↓	RO Col.↓	Stall↓	Rec. Time↓	R-FP↓	R-FN↓
Full	0.795	0.0093	0.0078	0.023	204.00	0.0540	0.0017
– Recoverability	0.795	0.0093	0.0078	0.023	204.00	0.0000	0.0000
– Counterfactual	0.763	0.0077	0.0083	0.022	230.25	0.0679	0.0014
– Progress shield	0.795	0.0093	0.0078	0.023	204.00	0.0540	0.0017
– Topology	0.790	0.0073	0.0071	0.028	210.58	0.0397	0.0009
Fixed formation	0.790	0.0073	0.0071	0.028	210.58	0.0397	0.0009

APPENDIX: REPRODUCIBILITY NOTES

- All learned methods are trained on the same generated dataset. For the current 500-episode seed used in the released run, both the shared learned-method dataset and the shared ablation dataset contain 59,381 graph samples.
- The heuristic teacher, rollout-score construction, topology-state updates, counterfactual selector, and optional shield are all expressed through shared environment-scale quantities rather than a scenario-specific list of tuned coefficients. The same deterministic rules are reused without per-scenario retuning.
- The main evaluation metrics are exactly those produced by the simulator:

$$\text{CollisionFree} = \mathbb{I}[\text{RRCol} = 0 \wedge \text{ROCol} = 0],$$

$$\text{FormOK} = \mathbb{I}[\text{FormRMS} < \epsilon_{\text{form}}],$$

$$\text{Deadlock} = \mathbb{I}[\text{stall_counter} \geq T_{\text{stall}} \wedge \neg \text{GoalReached}],$$

$$\text{Collapse} = \mathbb{I}[(\neg \text{CollisionFree} \wedge \neg \text{FormOK})$$

$$\vee \text{Deadlock} \vee (\chi_t > 0 \wedge b_t < \text{GoalProgress} \wedge \neg \text{FormOK})],$$

- while recoverability false positives / false negatives are per-step rates for predicting safe during collapse or predicting unsafe when collapse does not occur.
- The evaluation artifact stores additional metrics beyond the main paper tables, including topology switches, formation recovery time, and recoverability false-positive / false-negative rates; Tables VI and VII record the current single-seed dumps.
 - The repository also includes a zero-retune robustness suite and a derived-scale sensitivity suite. These drivers reevaluate fixed checkpoints under environment perturbations (obstacle density, LiDAR range, actuation limits, workspace scale, dynamic-obstacle speed) and under small perturbations of the shared geometric scales that feed topology and recoverability logic (spacing, sensing, clearance, horizon) without retraining.
 - The experiment driver runs the full artifact in the order: shared learned-method dataset generation, training of {GNN_ONLY, INSTANT_CERT, RVT_SWARM}, evaluation of those three learned methods plus {ADAPTIVE_FORMATION, CBF_QP_LIKE, ORCA_LIKE, CENTRALIZED_MPC}, shared ablation dataset generation, six ablation train/eval runs, and optional GIF visualization.
 - The ablation toggles are exactly: `full`, `no_recoverability`, `no_counterfactual`, `no_progress_shield`, `no_topology`, and `fixed_formation`; the last two are nearly degenerate once topology execution is disabled, which is why their gap should not be over-interpreted under single-seed runs.
 - The released visualization pipeline renders 28 per-method GIFs and 8 comparison GIFs for *open field* and *narrow passage* scenarios at team sizes 4 and 8, which is useful for qualitative diagnosis of topology switching behavior.